When does $\widehat{\beta}_{G\times\sigma^{KIS}} < 0$ mean labor is pulled into the informal sink?

A model-internal equivalence between the growth interaction and the between-sector flow

The estimate $\widehat{\beta}_{G\times\sigma^{KIS}} \approx -56.5$ (lower-middle income) is a coefficient in a *growth* regression; the "sucking into informal services" claim is about a *labor flow*. They are not the same object, and the coefficient alone does not establish the flow. What follows proves they are *equivalent inside the model*: given the structural assumptions below, the growth interaction is negative if and only if AI-induced reallocation moves labor into below-average-productivity sectors — which, under the sector ordering, is the informal sink τ.

Primitives and assumptions

Sectors $s \in \{a, m, \tau, k\}$ (agriculture, manufacturing, traditional/informal services, knowledge services), with employment shares θ_s, labor productivities p_s, and aggregate $P = \sum_s \theta_s p_s$. Growth is $g = \Delta P / P_{t-1}$.

Assumption 1 (McMillan–Rodrik accounting)**.** $\Delta P = \underbrace{\sum_s \theta_{s,t-1}\,\Delta p_s}_{W\ (within)} + \underbrace{\sum_s p_{s,t-1}\,\Delta\theta_s}_{B\ (between)}$.

Assumption 2 (Productivity response, U22)**.** *Holding F fixed, $\Delta p_s / p_{s,t-1} = g_s\, G + const$, where $g_s \equiv \alpha_s^G - \alpha_s^D \rho_s$ is the* net AI sensitivity *(boost minus displacement) of sector s.*

Assumption 3 (Reallocation, U23–U24)**.** *Target shares are the softmax $\theta_s^* \propto e^{\eta \ln p_s}$ and actual shares adjust as $\Delta\theta_s = \phi\,(\theta_s^* - \theta_{s,t-1})$ with speed $\phi = \phi(F, G) > 0$. Consequently (proved earlier) $\partial\theta_s^*/\partial G = \eta\,\theta_s^*(g_s - \bar{g})$, with $\bar{g} \equiv \sum_j \theta_j^* g_j = \langle g \rangle_{\theta^*}$.*

Assumption 4 (Absorptive gating)**.** *The realized KIS boost is gated by absorptive capacity: $\alpha_k^G \mapsto \alpha_k^G\,\psi(H, \sigma^{KIS})$ with ψ increasing and $\psi \to 0$ as $H, \sigma^{KIS} \to 0$. Hence in low-H, low-σ^{KIS} economies $g_k \to \alpha_k^G \psi - 0 \approx 0$ (the boost cannot land).*

Assumption 5 (Sector structure)**.** *Productivity ordering $p_k > p_m > p_a > p_\tau$ (τ is the low-productivity sink), and manufacturing is routine-intensive so ρ_m is large, making $g_m < 0$ strongly negative; the sink has low routine intensity and small boost, so $g_\tau \approx 0$.*

The central identity

Lemma 1 (The within channel cannot produce the negative)**.** $\dfrac{\partial W}{\partial G} = \sum_s \theta_{s,t-1}\, p_{s,t-1}\, g_s = \langle pg \rangle_\theta \;\geq$ 0 *whenever AI sensitivity is not concentrated in the lowest-productivity sectors. In particular, where σ^{KIS} measures* genuine *KIS (high p_k, high realized g_k), the within cross-partial with σ^{KIS} is* ***positive****. So any negative interaction must come from B.*

Proof. $W = \sum_s \theta_{s,t-1}\Delta p_s = \sum_s \theta_{s,t-1} p_{s,t-1} g_s G + \text{const}$ by Assumption 2; differentiate in G. Within holds employment fixed (no reallocation), so it is the mechanical productivity gain of incumbents and is signed by the p–g alignment, not by any flow. □

Lemma 2 (Between = covariance)**.** $\boxed{\dfrac{\partial B}{\partial G} = \phi\,\eta\ \mathrm{Cov}_{\theta^*}(p_s, g_s)}$ *where* $\mathrm{Cov}_{\theta^*}(p,g) = \langle pg\rangle_{\theta^*} - \langle p\rangle_{\theta^*}\langle g\rangle_{\theta^*}$.

Proof. By Assumptions 1,3, $\dfrac{\partial B}{\partial G} = \sum_s p_{s,t-1}\dfrac{\partial \Delta\theta_s}{\partial G} = \phi \sum_s p_{s,t-1}\dfrac{\partial \theta_s^*}{\partial G} = \phi\eta \sum_s p_{s,t-1}\,\theta_s^*\,(g_s - \bar{g})$. Since $\bar{g} = \langle g\rangle_{\theta^*}$, $\sum_s \theta_s^* p_s (g_s - \bar{g}) = \langle pg\rangle_{\theta^*} - \bar{g}\langle p\rangle_{\theta^*} = \mathrm{Cov}_{\theta^*}(p,g)$. □

Lemma 3 (Negative covariance ⇔ the sink gains)**.** $\dfrac{\partial B}{\partial G} < 0 \iff$ *the sectors gaining labor under AI are below-average productivity. Under Assumption 5 the gaining sector is the sink:* $\partial\Delta\theta_\tau/\partial G > 0$ *with* $p_\tau < \langle p\rangle_{\theta^*}$.

Proof. From Assumption 3, $\partial\Delta\theta_s/\partial G = \phi\eta\theta_s^*(g_s - \bar{g})$, so sector s *gains* labor iff $g_s > \bar{g}$ and *loses* iff $g_s < \bar{g}$. Labor therefore flows from low-g to high-g sectors. The between term $\partial B/\partial G = \phi\eta\,\mathrm{Cov}_{\theta^*}(p,g)$ is negative iff the high-g (gaining) sectors are the low-p ones — i.e. p and g are negatively aligned. Under Assumption 4, $g_k \to 0$ removes the high-p/high-g pairing; under Assumption 5, $g_m < 0$ drags $\bar{g}$ below $g_\tau \approx 0$, so $g_\tau > \bar{g}$ and the sink gains: $\partial\Delta\theta_\tau/\partial G > 0$. Because p_τ is the smallest productivity, this inflow makes $\mathrm{Cov}_{\theta^*}(p,g) < 0$. □

Theorem 1 (Equivalence)**.** *Under Assumptions 1–5, in a low-absorptive-capacity economy the AI growth effect operating through structural change is negative,*

$$\frac{\partial B}{\partial G} = \phi\eta\ \mathrm{Cov}_{\theta^*}(p,g) < 0,$$

if and only if *AI-induced reallocation pulls labor into the below-average-productivity informal sink,* $\partial\Delta\theta_\tau/\partial G > 0$ *with* $p_\tau < \langle p\rangle$. *Since (Lemma 1) the within channel is non-negative, a negative total interaction* $\widehat{\beta}_{G\times\sigma^{KIS}} < 0$ *is, within the model, equivalent to this inflow. The two statements — "AI lowers growth where services are large" and "AI pushes labor into informal services" — are the same fact, read off the between term.*

Regime	realized g_k	$\mathrm{Cov}_{\theta^*}(p,g)$	gaining sector	$\partial B/\partial G$
High base (H, σ^{KIS} high)	large	> 0	KIS (k), high p	> 0 (escape)
Low base (H, σ^{KIS} low)	≈ 0	< 0	sink (τ), low p	< 0 (trap)

Scope of the claim. This is a **model-internal** equivalence. It proves that the negative interaction is *exactly what the informal-inflow mechanism produces*, and that inside the model the coefficient and the flow are intertranslatable. It does *not* prove the empirical −56.5 was generated this way rather than by (i) σ^{KIS} mismeasuring informal bloat, (ii) AI destroying service jobs with no reallocation, or (iii) premature deindustrialization predating AI. To move from "equivalent in the model" to "demonstrated in the data," estimate the between term directly: compute $\sum_s p_{s,t-1}\Delta\theta_{st}$ for the lower-middle-income panel and test whether it is negative *and* carried by $p_\tau\,\Delta\theta_\tau$ with $\Delta\theta_\tau > 0$ — ideally with lagged G predicting subsequent $\Delta\theta_\tau$ to rule out (iii). That regression, with the informal employment share on the left-hand side, is what upgrades the claim from consistency to causation.